ENGINEERING DESIGN REPORT

LTE and 5G Architecture, Radio Access Design\nand Technology Selection

ECA56 – Advanced Wireless Communication Systems

CO2 Engineering System Architecture Evaluation Assignment

Report detail	Information
Course outcome	CO2 – Analyse LTE and 5G architectures, carrier aggregation, MIMO, beamforming and 5G Core service-based functions
Design scenario	Busy multi-use stadium; LTE continuity during migration to 5G NR and 5GC
Prepared by	P sridhar
Register number	1912260413
Submission date	11/11/2026

Executive Summary

This report proposes a phased LTE-Advanced and 5G deployment for a busy multi-use stadium. LTE service is retained through three-carrier aggregation while a 5G NR layer and service-based 5G Core are introduced. The supplied LTE component carriers provide 50 MHz aggregate bandwidth and 140 Mbit/s ideal downlink capacity; after the stated 20% overhead, estimated net throughput is 112 Mbit/s. For a 100 MHz NR carrier at 4 bit/s/Hz per stream, the ideal/net estimates are 400/300 Mbit/s for one stream, 800/600 Mbit/s for two streams and 1,600/1,200 Mbit/s for four streams.

A 64-element coherent array gives an ideal array gain of 18.06 dB; after the specified 3 dB implementation loss, the estimated gain is 15.06 dB. Hybrid beamforming is recommended for the 28 GHz stadium sector because it balances directional beam control and multiple simultaneous beams against the cost, power and RF-chain burden of a fully digital array. A CU/DU split is recommended, with time-critical lower-layer processing close to the stadium and higher-layer control centralized regionally. A local edge UPF is recommended for stadium applications that benefit from low latency and local breakout, with regional-core resilience and centralized services retained.

These values are planning estimates rather than guaranteed user speeds. Actual performance depends on scheduler sharing, propagation, device category, spatial channel rank, interference, event crowd distribution, backhaul and implementation losses. A controlled field trial is required before finalizing the radio and core design.

1. Requirements, Case Data and Design Assumptions

The operator must maintain LTE service during migration, add high-capacity service for dense crowds, introduce 5G NR at 28 GHz for the specified outdoor sector, and prepare the network for a service-based 5G Core. The report follows the six stages in the assignment and separates supplied numerical assumptions from design recommendations.

Design item	Supplied data
LTE-A component carriers	20 MHz at 1.8 GHz, efficiency 3 bit/s/Hz; 20 MHz at 2.1 GHz, efficiency 3 bit/s/Hz; 10 MHz at 2.6 GHz, efficiency 2 bit/s/Hz
LTE-A overhead	20% total throughput overhead after aggregation
5G NR carrier	100 MHz; representative efficiency 4 bit/s/Hz per spatial stream
NR spatial streams	Evaluate 1, 2 and 4 usable streams; equal efficiency per stream
NR overhead	25%
Beamforming sector	28 GHz outdoor sector; 64 array elements; 3 dB implementation loss
Radio architecture	LTE eNodeB/EPC; 5G gNodeB with configurable CU/DU split; RRC manages radio resource control and connection states
5G Core functions	AMF, SMF and UPF; compare local stadium-edge UPF with regional UPF

1.1 Assumptions and scope

- Throughput calculations use bandwidth × spectral efficiency. Decimal units are used: 1 MHz × 1 bit/s/Hz = 1 Mbit/s.
- The stated 20% and 25% overheads are applied multiplicatively to aggregate ideal throughput: net = ideal × (1 − overhead).
- The NR stream calculation assumes each spatial stream sustains the same 4 bit/s/Hz efficiency. This is an idealized comparison, not a guarantee that every user or cell can sustain four independent streams.
- The 64-element array gain follows the assignment formula 10 log10(N), and the 3 dB implementation loss is subtracted directly.
- A hybrid beamforming design is assumed to use multiple RF chains with analog phase-shifter networks; the exact number of RF chains and beams must be selected after hardware and traffic evaluation.

- The assignment does not specify stadium dimensions, event-user count, backhaul capacity, device mix, traffic profiles, link budget, channel model or required latency. These are treated as field-trial inputs, not invented values.

2. Stage 1 – LTE and 5G System Architecture Analysis

2.1 High-level LTE / LTE-Advanced architecture

In LTE, the UE connects over the LTE-Uu radio interface to an eNodeB. The eNodeB integrates radio resource management and important radio protocol functions and connects to the Evolved Packet Core (EPC). The EPC separates mobility/control functions from packet forwarding. A simplified control path is UE/eNodeB to the Mobility Management Entity (MME); a simplified user path is UE → eNodeB → Serving Gateway (S-GW) → Packet Data Network Gateway (P-GW) → external packet data network. The HSS stores subscriber information, while the PCRF provides policy and charging rules.

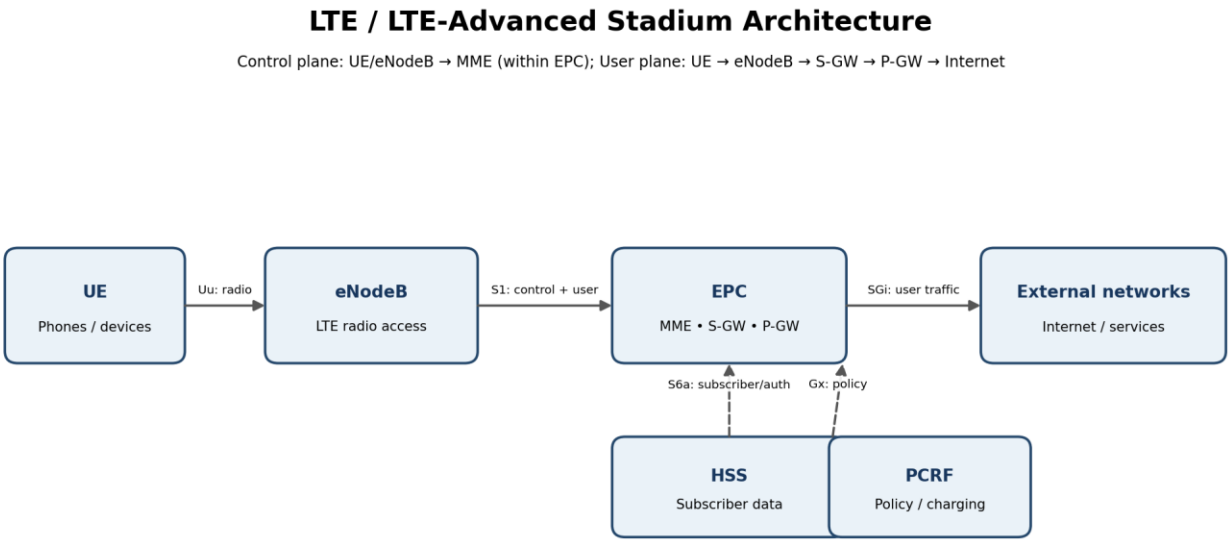


Figure 1. Simplified LTE/EPC architecture for the stadium. Solid arrows represent principal traffic or interface paths; dashed arrows represent control, subscriber or policy interactions.

2.2 High-level 5G NR / 5G Core architecture

In 5G, the UE connects through NR to a gNodeB. The gNodeB may be split into a Centralized Unit (CU) and one or more Distributed Units (DUs). The gNB connects to the 5G Core over N2 for control-plane signaling toward the AMF and N3 for user-plane traffic toward the UPF. The 5GC control plane contains network functions such as AMF and SMF and uses service-based interfaces for interactions among eligible network functions. The UPF forwards user packets toward a data network over N6.

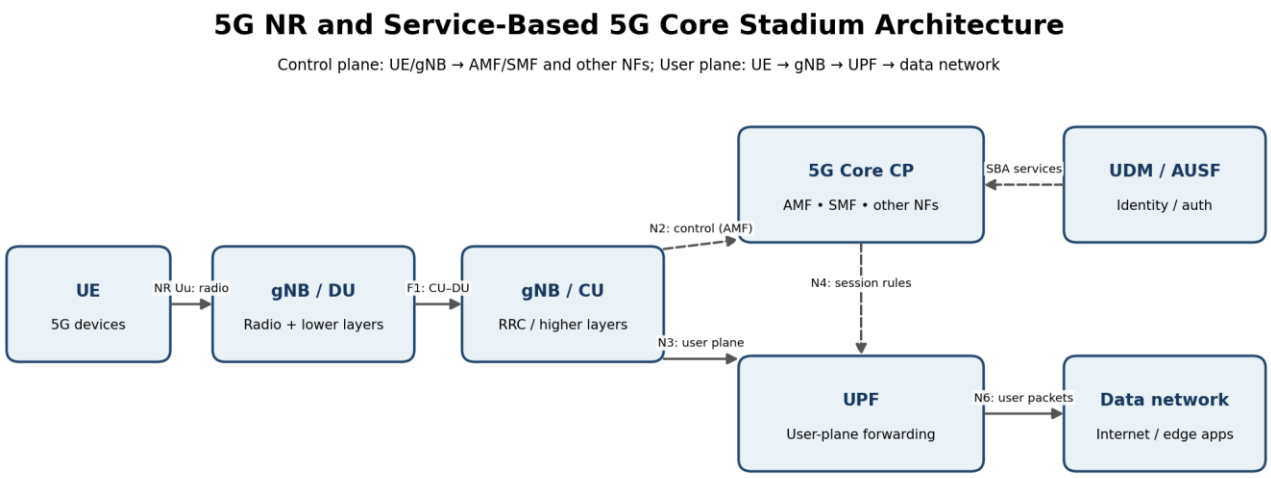


Figure 2. Simplified 5G NR and 5GC architecture. The diagram is a functional overview; detailed 5GC deployments contain additional network functions and security interfaces.

2.3 LTE versus 5G architecture comparison

Aspect	LTE / LTE-A	5G NR / 5GC	Stadium implication
Radio node	eNodeB integrates LTE radio access functions.	gNodeB; may be decomposed into CU and DU.	5G split allows distributed radio processing close to antennas and centralized higher-layer functions.
Core architecture	EPC with defined interfaces and gateway-based packet core.	Service-based 5GC control plane with separate UPF user plane.	Supports flexible deployment, edge breakout and modular scaling.
Control plane	MME handles mobility management and control signaling.	AMF handles access/mobility; SMF manages sessions; other NFs provide services.	Separate roles make control and session functions independently scalable.
User plane	eNodeB → S-GW → P-GW.	gNB → UPF → data network.	A local UPF can shorten the stadium application data path.
Service interaction	Primarily point-to-point reference interfaces between EPC entities.	SBA uses service-based interactions among 5GC network functions.	Supports modular functions and API-style service consumption, with security and lifecycle complexity.
Migration	Existing LTE devices and coverage remain supported.	New NR devices and 5GC capabilities are introduced.	Dual-generation operation avoids a disruptive cutover.

2.4 Service-Based Architecture (SBA)

The 5G SBA structures eligible control-plane network functions as service providers and service consumers. A network function can expose defined services that other authorized network functions discover and invoke through service-based interfaces. This modular approach supports independent scaling, software-based deployment, automation and flexible placement. It also introduces requirements for service registration/discovery, authentication and authorization, API security, observability, version compatibility and resilient cloud infrastructure. The user plane remains distinct: packet forwarding is performed by the UPF rather than by the SBA control-plane service interfaces.

3. Stage 2 – LTE-Advanced Carrier Aggregation Evaluation

Carrier aggregation (CA) combines multiple component carriers (CCs) so that a capable UE can use a wider aggregate radio resource. Component carriers may be in the same or different frequency bands. The network schedules

resources across configured carriers, while a primary component carrier supports essential control and mobility functions and secondary carriers add capacity.

Component carrier	Bandwidth	Spectral efficiency	Ideal contribution
CC1 – 1.8 GHz	20 MHz	3 bit/s/Hz	20 × 3 = 60 Mbit/s
CC2 – 2.1 GHz	20 MHz	3 bit/s/Hz	20 × 3 = 60 Mbit/s
CC3 – 2.6 GHz	10 MHz	2 bit/s/Hz	10 × 2 = 20 Mbit/s
Total	50 MHz	—	140 Mbit/s

Aggregate bandwidth

Btotal = 20 + 20 + 10 MHz

Result: Btotal = 50 MHz.

The UE can potentially access a 50 MHz aggregate allocation when all three carriers are configured, available and supported by the device and network.

Ideal downlink rate

Rideal = Σ(Bi × ηi) = (20 × 3) + (20 × 3) + (10 × 2)

Result: Rideal = 140 Mbit/s.

This is the sum of the component-carrier contributions before the specified overhead; it is not a per-user guaranteed throughput.

Estimated net LTE-A throughput

Rnet = 140 Mbit/s × (1 − 0.20)

Result: Rnet = 112 Mbit/s.

The 20% allowance represents the assignment's aggregate protocol/implementation overhead assumption. Actual user throughput will also depend on scheduling, load, retransmissions and radio conditions.

3.1 Practical effects of inter-band aggregation

- Coverage differences: 1.8 GHz and 2.1 GHz generally provide a broader coverage layer than 2.6 GHz under comparable antenna and power conditions. The 2.6 GHz carrier may be more sensitive to indoor penetration and cell-edge conditions.
- Scheduling: the scheduler must distribute traffic across component carriers according to radio quality, load, QoS and UE capability. Not every UE will use every carrier at every instant.
- Device capability: the UE must support the relevant band combination, bandwidth, RF chains and CA configuration. The network must check supported combinations.
- Control complexity: inter-band CA requires RF coordination, measurement configuration, mobility handling, power management and careful coexistence planning.
- Load and interference: adding carriers increases aggregate capacity but does not remove congestion if the backhaul, scheduler, control channels or available spatial resources become bottlenecks.

Decision: retain this three-carrier LTE-A configuration as a migration baseline. It offers a calculated 112 Mbit/s net sector/aggregate planning estimate under the supplied assumptions, preserves service for existing LTE users and provides a fallback layer while NR coverage and devices are introduced. It should not be interpreted as 112 Mbit/s for each stadium user; capacity is shared and varies with radio conditions and concurrent demand.

4. Stage 3 – 5G NR MIMO Strategies

Multiple-input multiple-output (MIMO) uses multiple transmit and receive antenna paths to exploit spatial diversity, spatial multiplexing and beamforming. With spatial multiplexing, independent data streams can be transmitted over sufficiently independent spatial channels. The assignment gives 4 bit/s/Hz per stream on a 100 MHz carrier.

Usable spatial streams	Ideal throughput calculation	Ideal throughput	Net after 25% overhead

1	100 MHz × 4 × 1	400 Mbit/s	300 Mbit/s
2	100 MHz × 4 × 2	800 Mbit/s	600 Mbit/s
4	100 MHz × 4 × 4	1,600 Mbit/s	1,200 Mbit/s

General NR throughput model

Rideal = B × η × Ns; Rnet = Rideal × 0.75

Result: 1 stream: 300 Mbit/s net; 2 streams: 600 Mbit/s net; 4 streams: 1,200 Mbit/s net.

The ideal model scales linearly with stream count because equal efficiency per stream is assumed. It excludes cell load sharing, scheduling gaps, retransmissions, control overhead beyond the stated allowance and user-specific channel limitations.

4.1 SU-MIMO and MU-MIMO comparison

Feature	SU-MIMO	MU-MIMO
Definition	Multiple spatial streams are sent to one UE.	Spatial resources are used to serve multiple UEs simultaneously.
Main gain	Raises peak throughput for a UE with multiple receive antennas and a high-rank channel.	Raises cell/sector capacity and can improve aggregate throughput by spatially separating users.
Requirements	UE antenna capability, favorable rank, adequate SINR and channel estimation.	Accurate channel state information, user separation, scheduling and interference control.
Limitation	A single UE may not support or benefit from all available streams.	Users with highly correlated channels or poor channel estimates may not be separable effectively.

4.2 Stadium recommendation and practical constraints

Recommend a flexible 5G NR MIMO design supporting up to four spatial layers for capable users in favorable channel conditions, combined with MU-MIMO scheduling to serve multiple users across the dense stadium crowd. SU-MIMO supports high-throughput users, while MU-MIMO improves aggregate sector capacity when users have separable spatial signatures. The system should adapt the number of layers to measured channel rank and UE capability rather than forcing four streams for every user.

- Channel rank and propagation: rich scattering can support multiple independent paths, but line-of-sight dominance, correlated antennas, blockage and polarization effects can reduce rank.
- UE capability: many handsets have fewer receive chains than the base-station array; supported layers vary by device and band.
- CSI quality and mobility: stale or inaccurate channel-state information reduces precoding accuracy, especially when users move or channels change quickly.
- Interference and scheduler limits: simultaneous streams can interfere, and scheduler fairness/QoS constraints may prevent all users from receiving maximum spatial resources.
- Hardware and thermal limits: RF chains, power amplifiers, calibration, cooling and device battery consumption constrain sustained operation.

The four-stream net figure of 1.2 Gbit/s is therefore an idealized capacity estimate for the stated carrier and efficiency assumptions, not a prediction of a single user's real application throughput.

5. Stage 4 – Beamforming Strategy Selection

Ideal coherent array gain for 64 elements

Gideal = 10 log10(N) = 10 log10(64)

Result: Gideal = 18.06 dB.

The formula is the assignment's ideal coherent array-gain model. It assumes element signals are phase-aligned toward the intended direction.

Estimated array gain after implementation loss

Gestimated = 18.06 dB − 3 dB

Result: Gestimated = 15.06 dB.

This is the net gain estimate after the stated implementation loss. It is not the total antenna gain or complete link budget; element gain, feeder loss, EIRP limits and propagation must also be included.

5.1 Analog, digital and hybrid beamforming

Criterion	Analog beamforming	Digital beamforming	Hybrid beamforming
Beam control	Phase shifters form one/few beams; limited simultaneous independent beams.	Per-element digital weighting gives fine-grained beam and null control.	Analog subarrays plus digital processing provide multiple controllable beams.
RF chains	Typically few RF chains, often one per analog beam.	Many RF chains, potentially one per antenna element.	Fewer RF chains than antenna elements; one per subarray/stream as designed.
Streams / users	Limited spatial multiplexing flexibility.	Best flexibility for multi-user and multi-stream precoding.	Supports multiple beams/streams, limited by RF-chain count.
Power and cost	Lower RF-chain power and hardware cost.	Highest RF-chain, converter, processing and power burden.	Intermediate cost, power and complexity.
Implementation	Simpler RF architecture but beam control is constrained.	Complex calibration, synchronization and baseband processing.	More complex than analog but less demanding than fully digital.

5.2 Selected approach: hybrid beamforming at 28 GHz

Hybrid beamforming is recommended for the stadium's 28 GHz outdoor sector. At mmWave frequencies, compact antenna arrays can form narrow directional beams that compensate for higher path loss and concentrate energy toward users. Hybrid architecture uses analog phase shifting within antenna subarrays and digital processing across a smaller number of RF chains. It provides more than one controllable beam and supports spatial multiplexing without the cost and power of a fully digital RF chain for every one of the 64 elements.

- Beamsteering tracks the direction of a UE and maintains a directional link as the user moves through the stadium perimeter or concourse.
- Spatially separated beams can serve different users or zones and reduce energy radiated toward unintended directions.
- Narrow beams can improve link margin and spatial reuse, but beam management and rapid recovery from blockage are essential.

Hybrid beamforming may perform poorly when users move rapidly, beams are frequently blocked by people or vehicles, channel estimates become stale, the RF-chain count is too small for the requested simultaneous streams, or phase-shifter quantization/calibration errors are significant. A robust deployment needs beam sweeping, beam-failure recovery, alternate beams and lower-frequency LTE/NR coverage as fallback. The 15.06 dB array-gain estimate must be validated with the actual array pattern and EIRP limits.

6. Stage 5 – 5G RAN Design and Functional Entities

6.1 Proposed stadium gNodeB arrangement

Deploy a distributed set of 28 GHz outdoor radio units/antenna arrays around suitable stadium sectors and high-demand outdoor approaches. Connect them to local DUs located close enough to meet fronthaul timing and capacity requirements. Connect the DUs to a regional or campus-edge CU over a high-capacity, low-latency

transport network. Retain LTE eNodeBs and EPC connectivity during migration, with interworking and mobility procedures configured for LTE/NR coverage continuity.

Entity	Role in the stadium design
gNodeB (gNB)	5G NR base-station system providing radio access, scheduling, radio protocols and connectivity to the 5GC. It may be implemented as CU and DU functions.
Centralized Unit (CU)	Hosts higher-layer functions, commonly including RRC and PDCP-related functions depending on the selected split. Supports centralized coordination and mobility/control functions.
Distributed Unit (DU)	Hosts lower-layer real-time radio functions, commonly including RLC, MAC and high-PHY functions depending on the split. Located near the radio site to meet timing constraints.
RRC	Radio Resource Control manages connection establishment, reconfiguration and release; broadcasts system information; configures measurements, mobility and radio bearers. RRC is a radio control-plane protocol, not a 5GC network function.
Radio unit / antenna array	Performs RF transmission/reception and beamforming functions; connects to DU according to the chosen implementation.

6.2 Recommended functional split

Use a CU/DU split with lower-layer and timing-sensitive processing at the stadium-side DU and higher-layer control functions centralized at an edge/regional CU. This balances real-time response and efficient pooling. It is important not to treat the split as a single universal boundary: exact protocol placement depends on the chosen 3GPP split option and vendor implementation.

Design choice	Benefits	Trade-offs
CU centralized; DU local	Centralizes higher-layer functions, simplifies multi-site coordination and software management, while local DU handles time-sensitive lower layers.	Requires reliable low-latency transport between DU and CU; CU failure or transport congestion can affect multiple cells.
More integrated gNB at site	Reduces dependence on fronthaul for split processing, can simplify latency and local resilience.	Less pooling and centralized coordination; higher equipment footprint at each site and potentially less efficient scaling.

The recommended split requires high-capacity, deterministic fronthaul and synchronization. If the transport cannot satisfy latency, jitter, availability and throughput requirements, move more functions closer to the site or use a more integrated gNB. RRC remains part of the gNB radio control architecture and coordinates UE radio connection states; AMF handles core access and mobility functions. These roles must not be confused.

7. Stage 6 – 5G Core Procedures and Design Decision

7.1 AMF, SMF and UPF responsibilities

5GC function	Primary responsibility	In the stadium scenario
AMF – Access and Mobility Management Function	Registration, connection management, reachability and mobility/access signaling; terminates relevant NAS signaling.	Handles UE registration and access/mobility control; coordinates with other core functions for authentication and policy-related procedures.
SMF – Session Management Function	Establishes, modifies and releases PDU sessions; selects and controls UPF; manages session-related IP addressing and forwarding rules.	Creates the UE data session and configures the selected UPF through the N4 interface.
UPF – User Plane Function	Forwards user packets, applies forwarding/QoS rules and connects the 5G access network to data networks.	Carries stadium user traffic to local edge applications or onward to regional/external networks.

7.2 Representative UE registration flow

- The UE detects a suitable NR cell and establishes radio access signaling with the gNB; RRC connection procedures establish the radio control connection.
- The UE sends a NAS Registration Request through the gNB toward the AMF. The gNB carries the control signaling over the N2 interface.
- The AMF selects or interacts with the appropriate authentication/subscriber functions (for example, AUSF and UDM) to authenticate the subscriber and obtain subscription data. These functions are included here to explain the functional path even though the assignment focuses on AMF, SMF and UPF.
- The AMF manages registration and mobility context, coordinates security and registration procedures, and returns the appropriate registration outcome through the gNB to the UE.
- After registration, the UE is in a state where it can request a PDU session for data connectivity.

7.3 Representative PDU session and user-data flow

- The UE sends a PDU Session Establishment Request via the gNB to the AMF.
- The AMF selects or contacts an SMF based on subscription, requested data network and policy context.
- The SMF selects an appropriate UPF and establishes session/forwarding rules over the N4 interface (using PFCP control).
- The gNB and UPF user-plane path is established through N3. The gNB receives the tunnel and QoS-related configuration needed to forward user packets.
- User packets then flow UE → gNB → UPF → data network over N6. Return packets follow the reverse user-plane path.
- The AMF and SMF retain control-plane responsibilities while the UPF forwards the data packets; this is the practical control-plane/user-plane separation.

7.4 Local stadium UPF versus regional UPF

Criterion	Local stadium-edge UPF	Regional UPF
User-plane path	Short path from gNB to nearby UPF and local breakout to stadium applications.	Traffic travels from stadium RAN to a more distant regional core location before reaching data networks.
Latency	Can reduce transport distance and delay for local applications, subject to edge compute and transport design.	Usually adds transport distance; may be adequate for ordinary internet and non-latency-critical services.
Traffic routing	Supports local traffic breakout, local content, venue services and selected enterprise applications.	Centralizes routing and policy enforcement; simpler for common regional services.
Resilience / operations	Requires local power, compute, security, redundancy, monitoring and skilled support.	Can benefit from regional pooling and centralized operations; dependency on regional transport path increases.
Scale and cost	Additional distributed infrastructure and lifecycle cost.	Potentially more efficient resource pooling and fewer edge locations.

Recommendation: deploy a local UPF at the stadium edge for venue applications, local content, event operations and traffic that benefits from short user-plane paths. Retain regional UPF capability for general internet traffic, centralized services, resilience and sessions that do not require local breakout. The SMF selects and programs the appropriate UPF based on policy, DNN, slice/service needs and topology. This is a selective edge model rather than a requirement to anchor every session locally.

7.5 Control-plane and user-plane separation; SBA

Control-plane functions manage registration, mobility, authentication coordination, policy and PDU session setup. User-plane functions forward application packets. Separating these planes permits control functions to scale independently from packet-forwarding capacity and enables user traffic to be placed closer to applications. The SBA supports communication among 5GC control-plane network functions through defined services and

discovery/authorization mechanisms. It does not mean user packets traverse the service-based APIs; packet forwarding remains on the UPF user plane.

8. Integrated Stadium Migration Recommendation

Phase	Deployment action	Reason and expected outcome
1 – Maintain LTE continuity	Keep the existing LTE eNodeB/EPC service active; configure the three LTE-A component carriers for capable devices.	Calculated aggregate bandwidth is 50 MHz; ideal rate is 140 Mbit/s and estimated net rate is 112 Mbit/s after 20% overhead. Protects legacy devices and provides broad-area fallback.
2 – Introduce 5G NR coverage	Deploy 100 MHz NR capacity where spectrum and equipment permit; begin with 1–2 spatial layers and enable up to 4 where radio conditions and UE support allow.	NR estimates are 300, 600 and 1,200 Mbit/s net for 1, 2 and 4 streams respectively after 25% overhead. Adaptive layers avoid assuming ideal rank everywhere.
3 – Add 28 GHz hybrid beamforming	Install 64-element hybrid arrays on outdoor sectors with beam management and blockage recovery.	Estimated array gain after implementation loss is 15.06 dB. Hybrid balances capacity, beam agility, RF-chain count and power.
4 – Split RAN functions	Use local DUs near the radio sites and a centralized edge/regional CU, provided fronthaul meets timing and availability requirements.	Local lower-layer processing supports real-time radio functions; centralized higher layers support coordination and operational pooling.
5 – Deploy 5GC with edge breakout	Introduce AMF/SMF control functions and a local stadium UPF for local applications, with regional UPF routing/resilience for other services.	Reduces user-plane path length for selected applications while preserving centralized service options and operational resilience.
6 – Optimize and scale	Use event-day telemetry, QoS, beam/sector load, CA use, MIMO rank and handover data to tune capacity and add radios where needed.	Makes expansion evidence-based and prevents overbuilding from idealized throughput calculations.

8.1 Field-trial verification plan

- Throughput and load: test each LTE component carrier, aggregate CA performance, NR bandwidth and net throughput at multiple crowd densities and times. Compare measured rates with 112 Mbit/s LTE-A and the 300/600/1,200 Mbit/s NR planning estimates while accounting for per-user sharing.
- UE capability: inventory representative handsets and verify band combinations, CA support, NR bands, MIMO layer capability and antenna behavior.
- MIMO performance: measure rank indicator, channel quality, layer utilization, SU-MIMO and MU-MIMO scheduling gain, fairness and retransmissions.
- Beamforming: measure array pattern, realized gain, EIRP, beam acquisition/recovery time, blockage impact, beam switching and coverage around stadium entrances and seating/outdoor areas.
- RAN split/transport: verify CU–DU latency, jitter, fronthaul capacity, synchronization, packet loss, failover and behavior during transport congestion.
- 5GC and UPF: trace registration and PDU session success, AMF/SMF signaling latency, UPF throughput, local-breakout routing, regional fallback, security and redundancy.
- Mobility and continuity: test LTE-to-NR mobility, NR beam transitions, congestion, session continuity and fallback during the event crowd movement.
- Operational readiness: validate power, cooling, site access, monitoring, cybersecurity, software lifecycle, emergency procedures and support staffing.

9. Conclusion

The recommended stadium design is a phased multi-layer migration that preserves LTE while adding NR capacity and a service-based 5G Core. The LTE-A carrier aggregation baseline combines 50 MHz across 1.8, 2.1 and 2.6 GHz and

produces an estimated 112 Mbit/s net planning rate. The 100 MHz NR carrier yields estimated net rates of 300, 600 and 1,200 Mbit/s for one, two and four spatial streams under the assignment's equal-efficiency assumption. These values demonstrate the potential gain from bandwidth and spatial multiplexing, but actual user rates depend on load, radio conditions and device capability.

For the 28 GHz sector, the 64-element array provides an estimated 15.06 dB gain after the specified implementation loss. Hybrid beamforming is the preferred compromise for multiple directional beams with manageable RF-chain count, power and complexity. A CU/DU split with local DU processing and centralized higher-layer functions is recommended when transport meets the required performance. In the core, AMF manages access and mobility, SMF manages sessions and UPF forwards user traffic. A local stadium UPF should serve latency-sensitive venue applications, while regional UPF services provide centralized connectivity and resilience.

The final design should be accepted only after a field trial verifies real throughput under crowd load, UE capabilities, MIMO rank, beam robustness, fronthaul performance, mobility, local breakout, security and operational resilience. The report's calculations are transparent planning estimates based on the assignment data and should not be interpreted as guaranteed commercial performance.

References

- [1] 3GPP TS 23.501, System Architecture for the 5G System (5GS); Stage 2.
- [2] 3GPP TS 23.502, Procedures for the 5G System (5GS); Stage 2.
- [3] 3GPP TS 38.300, NR and NG-RAN Overall Description.
- [4] 3GPP TS 36.300, E-UTRA and E-UTRAN; Overall Description.
- [5] 3GPP TS 38.401, NG-RAN; Architecture Description.
- [6] 3GPP TS 38.331, NR; Radio Resource Control (RRC) Protocol Specification.
- [7] ECA56 Advanced Wireless Communication Systems, CO2 Engineering System Architecture Evaluation Assignment, supplied planning data and requirements.

Note: The assignment-provided spectral efficiencies and overheads are treated as assessment planning assumptions, not fixed 3GPP performance guarantees. Architecture diagrams are high-level functional representations; detailed interfaces and function placement depend on the selected release, deployment option and vendor implementation.

Appendix A – Key Results Summary

Calculation	Result
LTE-A aggregate bandwidth	50 MHz
LTE-A ideal throughput	140 Mbit/s
LTE-A net throughput after 20% overhead	112 Mbit/s
NR 1 stream ideal / net	400 / 300 Mbit/s
NR 2 streams ideal / net	800 / 600 Mbit/s
NR 4 streams ideal / net	1,600 / 1,200 Mbit/s
64-element ideal array gain	18.06 dB
64-element gain after 3 dB loss	15.06 dB
Recommended beamforming	Hybrid beamforming
Recommended RAN split	Local DU + centralized edge/regional CU, subject to fronthaul validation
Recommended UPF placement	Local stadium UPF for local/low-latency applications plus regional UPF for general services and resilience